OMER GADALLA
My Portfolio

Galaxy Rocket Team

My passion for space began at a young age and inspired me to pursue a degree in Aerospace Engineering. Among all the subjects I studied, I've always been most drawn to rocket propulsion and space systems. In 2025, I joined the Galaxy Rocket Team, working towards competing in the TEKNOFEST Rocket Competitions. Through this experience, I gained hands-on expertise in **rocket design, Bipropellant rocket engine design and system engineering** while collaborating closely with a multidisciplinary team. Balancing this project alongside METU's hard coursework strengthened both my technical foundation and my ability to perform under pressure, preparing me to contribute effectively to innovative companies.

Project 1: TEKNOFEST Liquid Bipropellant Rocket Engine Development Competition (B3) (2025–2026)

Situation: TEKNOFEST Liquid Bipropellant Rocket Engine competition challenged teams to design and test a fully functional LOX/Jet-A1 engine over two years.

Task : Work with a 15-member team with our professor as an advisor to design a **100 N, 10-second, 3000 kelvin,** liquid rocket engine meeting safety and performance requirements

My Actions in the project:

Position: Propulsion Analysis Team Lead

- Performed **combustion** and **CFD analysis** in ANSYS Fluent.
- Conducted **thermal and structural analysis** to ensure chamber durability under 10 bar.
- Designed and optimized the **cooling system** for the combustion chamber.
- Developed **thermo-structural** analysis to validate material strength under extreme conditions.

Figure 1 The 100N Rocket Engine

Gained Experience:

- **Combustion chamber** design, analysis, manufacturing and testing.
- Knowledge of **Cryogenics** and **Hypergolic** fuels.
- Knowledge of **Bill of Materials** (BOM).
- Familiarity with **instrumentation** and control components.

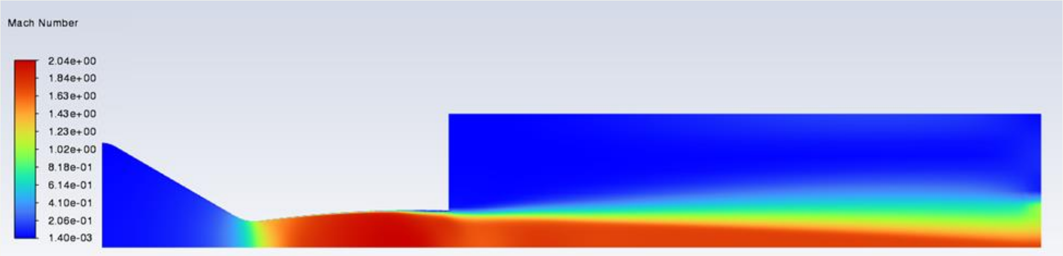


Figure 3 Nozzle CFD

Figure 2 The 10-bar Combustion Chamber

Project 2: TEKNOFEST High Altitude Rocket Competition (2025)

Situation: For the TEKNOFEST rocket competition, our team was tasked with designing a high-altitude rocket capable of reaching 15,000 ft, deploying a parachute, and ensuring the rest of the rocket landed safely without any damage to its subsystems.

Task: We designed and built a 2-meter rocket capable of reaching Mach 1.4, soaring to 15,000 ft and landing safely.

My Actions in the project:

Position: CFD & Structural Analysis Team Lead

- Lead **aerodynamic** and **structural** design, ensuring performance and safety under high loads.
- Conducted **CFD** simulations in ANSYS Fluent to calculate drag at peak conditions.
- Imported results into ANSYS Mechanical for **structural** and **buckling** analysis.
- Selected **Aluminum 7075** for motor block based on load requirements.
- Integrated composites (carbon fiber, glass fiber, aluminum) to balance weight and strength.

Result: Produced an **optimized rocket structure** with validated aerodynamic and structural safety margins.

Figure 5 The High-Altitude Rocket

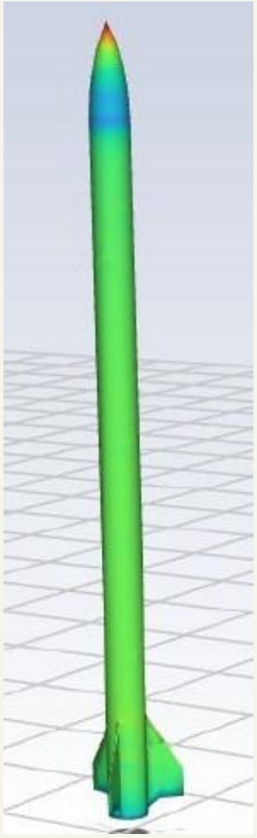


Figure 4 Rocket CFD

Gained Experience

- Hands-on expertise in rocket design projects.
- External aerodynamic **CFD analysis**.
- **Composite structure** design and analysis.
- Structural analysis.

<u>**Project 3: Sustainable 20N Bipropellant Thruster**</u>

Situation: During my internship at MTS Defense, the company aimed to enter Türkiye's growing space industry.

Task: Design a cost-effective, green-propellant thruster for small satellites (10–100 kg) to showcase feasibility and innovation.

Action:

R&D Engineer

- Arranged a roadmap to manage development and follow progress.
- Conducted R&D on propulsion subsystems and mission requirements.
- Performed trade-off analysis of H_2O_2/Kerosene vs. N_2O/Propylene, selecting N_2O/Propylene for safety and sustainability.
- Designed CAD model, P&ID, and set chamber pressure (10 bar) and thrust target (20 N).

Result: Delivered a complete conceptual design package that demonstrated the company's potential entry into the smallsat propulsion market and aligned with sustainable space initiatives.

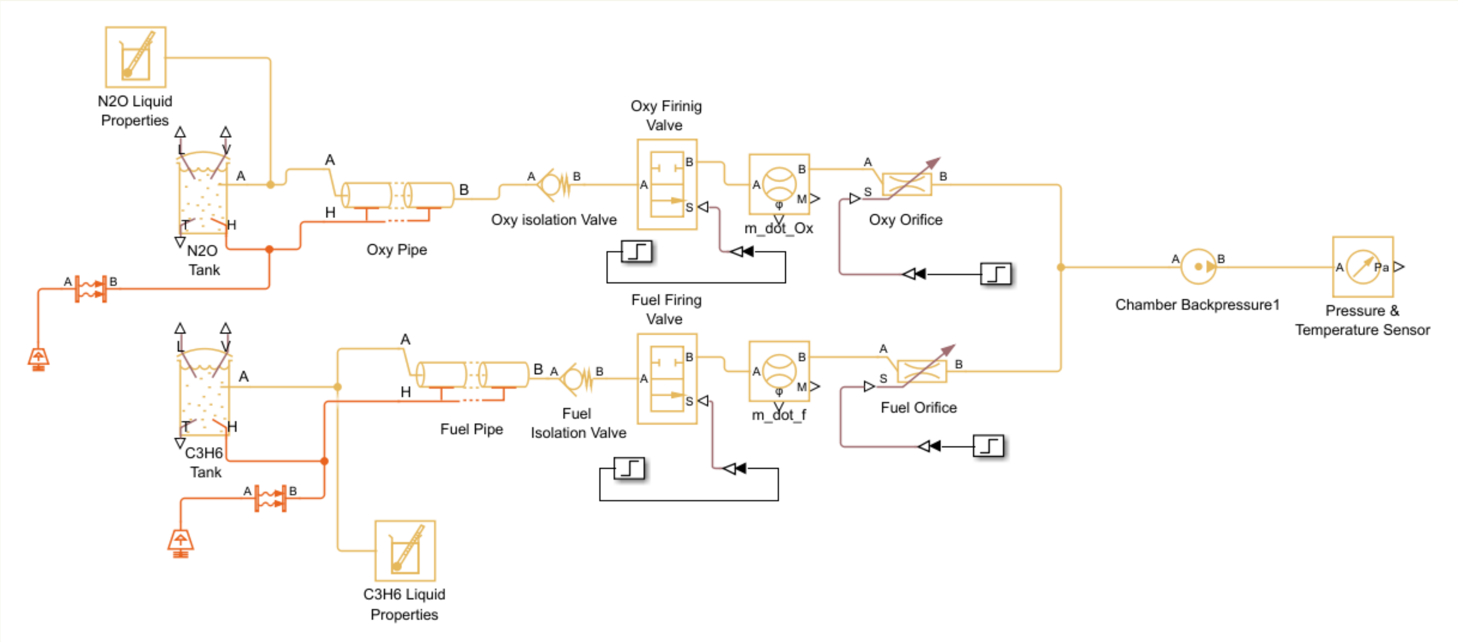


Figure 6 20N Bipropellant Thruster Simscape Model

<u>**Gained Experience:**</u>

- Knowledge of **sustainable aviation fuels.**
- Design and production of propellants and **gas systems.**
- Familiarity with **spacecraft subsystems**.
- Simulation and Modeling of satellite **propulsion subsystems** in **Simscape**.
- Familiarity with **instrumentation** and control components.

Figure 7 3D-Printed Nozzle prototype in hands

METU Projects

Project 4: CFD Analysis of a Converging-Diverging Nozzle (ASE 342 – Aerodynamics II)

Situation: As part of Aerodynamics II coursework, a CFD analysis of a nozzle flow was assigned.

Task: Validate CFD models against experimental data while reducing computational cost.

Action:

- Choose the nozzle geometry based on NASA's published experiment.
- Built viscous compressible flow simulations in **ANSYS Fluent** with **boundary-layer** capturing.
- Implemented axisymmetric simplification to cut computational time while preserving accuracy.

Result: Achieved **CFD** predictions within 5–10% deviation from NASA data and quantified thrust across flow regimes, proving model reliability.

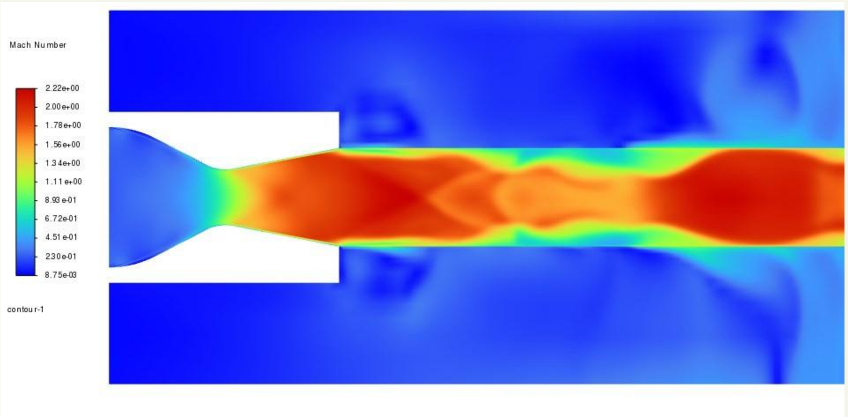
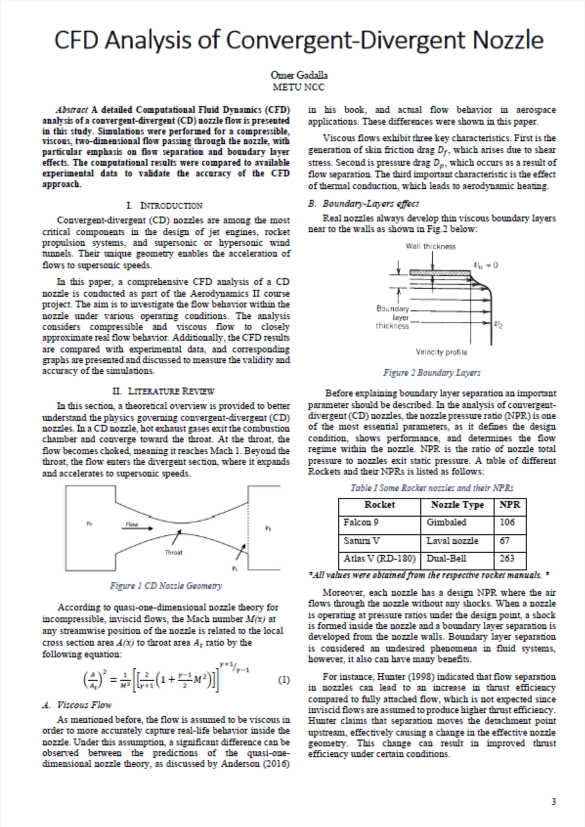


Figure 8 Nozzle CFD Results



CFD Analysis of Convergent-Divergent Nozzle

Omer Gadalla
METU NCC

Abstract **A detailed Computational Fluid Dynamics (CFD) analysis of a convergent-divergent (CD) nozzle flow is presented in this study. Simulations were performed for a compressible, viscous, two-dimensional flow passing through the nozzle, with particular emphasis on flow separation and boundary layer effects. The computational results were compared to available experimental data to validate the accuracy of the CFD approach.**

I. Introduction

Convergent-divergent (CD) nozzles are among the most critical components in the design of jet engines, rocket propulsion systems, and supersonic or hypersonic wind tunnels. Their unique geometry enables the acceleration of flows to supersonic speeds.

In this paper, a comprehensive CFD analysis of a CD nozzle is conducted as part of the Aerodynamics II course project. The aim is to investigate the flow behavior within the nozzle under various operating conditions. The analysis considers compressible and viscous flow to closely approximate real flow behavior. Additionally, the CFD results are compared with experimental data, and corresponding graphs are presented and discussed to measure the validity and accuracy of the simulations.

II. Literature Review

In this section, a theoretical overview is provided to better understand the physics governing convergent-divergent (CD) nozzles. In a CD nozzle, hot exhaust gases exit the combustion chamber and converge toward the throat. At the throat, the flow becomes choked, meaning it reaches Mach 1. Beyond the throat, the flow enters the divergent section, where it expands and accelerates to supersonic speeds.

Flow
Throat

Figure 1 CD Nozzle Geometry

According to quasi-one-dimensional nozzle theory for incompressible, inviscid flows, the Mach number *M(x)* at any streamwise position of the nozzle is related to the local cross section area *A(x)* to throat area A_t ratio by the following equation:

$$\left(\frac{A}{A_t}\right)^2 = \frac{1}{M^2}\left[\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]\right]^{\gamma+1/\gamma-1} \quad (1)$$

A. Viscous Flow

As mentioned before, the flow is assumed to be viscous in order to more accurately capture real-life behavior inside the nozzle. Under this assumption, a significant difference can be observed between the predictions of the quasi-one-dimensional nozzle theory, as discussed by Anderson (2016) in his book, and actual flow behavior in aerospace applications. These differences were shown in this paper.

Viscous flows exhibit three key characteristics. First is the generation of skin friction drag D_f, which arises due to shear stress. Second is pressure drag D_p, which occurs as a result of flow separation. The third important characteristic is the effect of thermal conduction, which leads to aerodynamic heating.

B. Boundary-Layers effect

Real nozzles always develop thin viscous boundary layers near to the walls as shown in Fig.2 below:

Wall thickness
$v_w = 0$
Boundary layer thickness
v_2
Velocity profile

Figure 2 Boundary Layers

Before explaining boundary layer separation an important parameter should be described. In the analysis of convergent-divergent (CD) nozzles, the nozzle pressure ratio (NPR) is one of the most essential parameters, as it defines the design condition, shows performance, and determines the flow regime within the nozzle. NPR is the ratio of nozzle total pressure to nozzles exit static pressure. A table of different Rockets and their NPRs is listed as follows:

Table I Some Rocket nozzles and their NPRs

Rocket	Nozzle Type	NPR
Falcon 9	Gimbaled	106
Saturn V	Laval nozzle	67
Atlas V (RD-180)	Dual-Bell	263

****All values were obtained from the respective rocket manuals. ****

Moreover, each nozzle has a design NPR where the air flows through the nozzle without any shocks. When a nozzle is operating at pressure ratios under the design point, a shock is formed inside the nozzle and a boundary layer separation is developed from the nozzle walls. Boundary layer separation is considered an undesired phenomena in fluid systems, however, it also can have many benefits.

For instance, Hunter (1998) indicated that flow separation in nozzles can lead to an increase in thrust efficiency compared to fully attached flow, which is not expected since inviscid flows are assumed to produce higher thrust efficiency. Hunter claims that separation moves the detachment point upstream, effectively causing a change in the effective nozzle geometry. This change can result in improved thrust efficiency under certain conditions.

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Figure 9 My Academic Paper

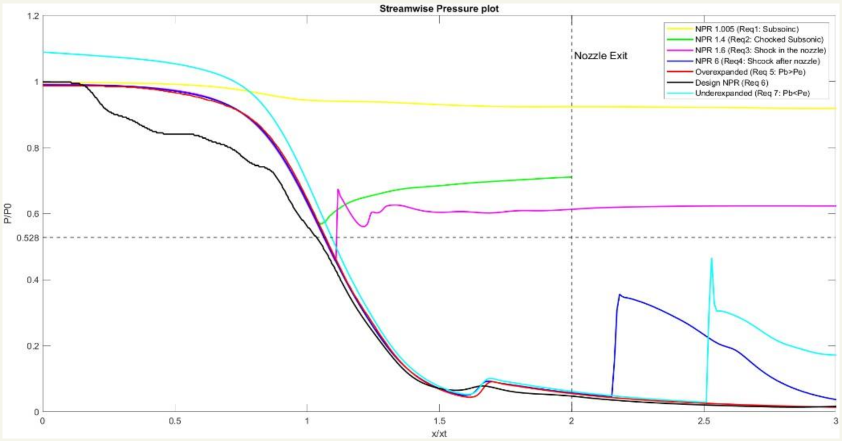

Figure 10 Pressure Distribution Across the Nozzle

Gained Experience

- Acquired hands-on expertise in **CFD simulations**.
- Gained in-depth knowledge of **internal viscous compressible flows** under **high-speed** and **high-pressure** conditions.
- Strengthened understanding of **real-world aerodynamics** through data validation with NASA's experiment.
- Enhanced **technical report writing** and scientific communication skills by preparing detailed documentation of findings.

Project 5: Rocket Nozzle Design (ASE 435 – Propulsion Systems II)

Situation: Tasked with designing a supersonic nozzle using the Method of Characteristics (MOC).

Task: Develop a MATLAB tool to generate an optimized nozzle contour for Mach 3 exit conditions

Action:

- Implemented MOC for expansion flow fields
- Created Prandtl-Meyer function solver for precise Mach number calculation.

Result: Produced an **optimized nozzle contour** with smooth supersonic expansion, demonstrating practical application of MOC in propulsion design.

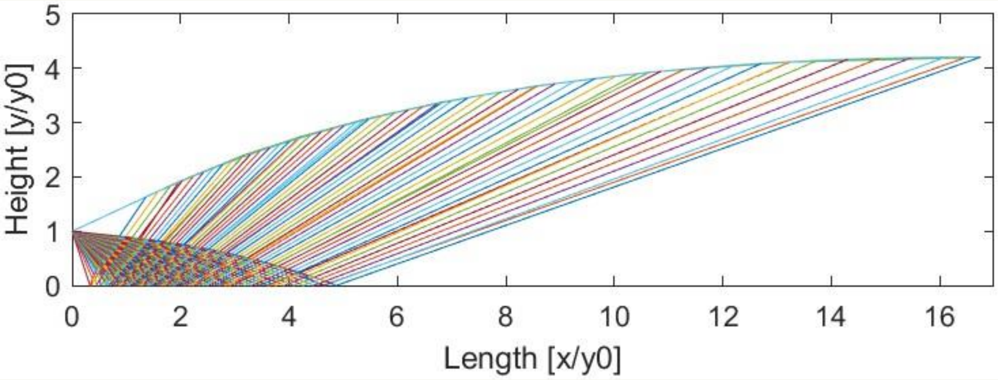

Figure 11 Rocket Nozzle Contour

Project 6: Combustion Analysis of Rocket Combustion (ASE 492 - Rocket Technology)

Situation: The course required simulating equilibrium combustion for a hydrogen-oxygen engine.

Task: Develop a MATLAB-based solver to calculate equilibrium mole fractions of combustion species.

Action:

- Implemented **Newton-Raphson** method and MATLAB's **fsolve** for chemical equilibrium equations.
- Modeled thermodynamic properties for six combustion species.

Results: Created a validated tool that deepened understanding of reaction kinetics and thermodynamics in rocket engines.

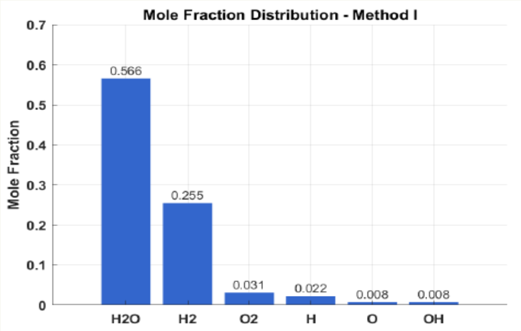

Figure 13 Analysis Results – method I

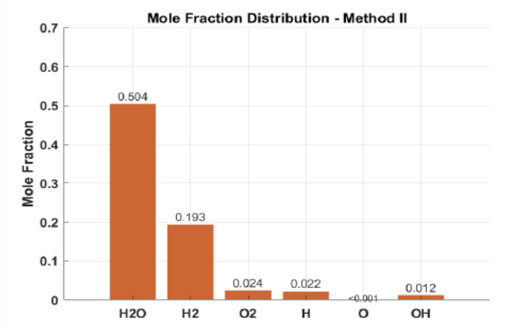

Figure 12 Analysis Results – method II

Gained Experience

- **MATLAB** code writing.
- Knowledge of the **Method of Characteristics.**
- Knowledge of rocket **nozzle optimization.**
- Practical knowledge of **supersonic flow** and **expansion waves**.
- Real-life rocket **combustion analysis**.
- Knowledge of **Numerical Methods**.

My Certifications

MATHWORKS. MATLAB Onramp

MATHWORKS. Simscape Onramp

Rolls-Royce. Reliability and Safety Workshop

Figes A.Ş. MATLAB & Simulink

Udemy. Aerospace Structures and Materials

Udemy. Rocket Science and Engineering